

BAULKHAM HILLS HIGH SCHOOL

HALF YEARLY EXAMINATION

2009

YEAR 11

# MATHEMATICS

## Extension 1

Time Allowed:

90 minutes  
(plus 5 mins reading time)

**Instructions:**

- Write on the paper provided
- Do not write on this question paper
- Show all working
- Use black or blue pens only
- Start a new page for each question
- Write your name and your teacher's name at the top of each page
- Board approved calculators may be used

**QUESTION 1      Start on a new page****Marks**

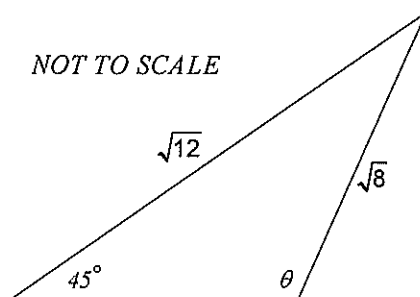
- a) Solve  $(x+4)^2 = 3(x+4)$  2
- b) Solve  $\frac{1}{|x+5|} \geq 1$  2
- c) Find the largest possible domain of 3  
$$f(x) = \frac{\sqrt{9-x^2}}{\sqrt{x+1}}$$
- d) Solve  $4^{2x+5} = 8$  2
- e) State the range of  $f(x) = \frac{|x|}{\sqrt{x^2+4}}$  2

**QUESTION 2      Start on a new page**

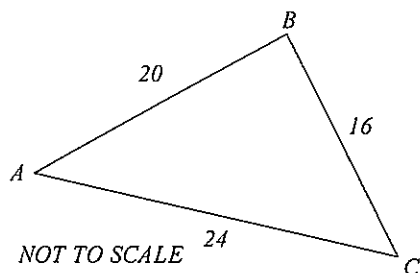
- a) If  $2x + y + 3 + \sqrt{4x-3y} = 5 + 2\sqrt{6}$ , find the value of  $x$  and  $y$  3
- b) Is the function  $f(x) = x^3 \cos x$  odd, even or neither? Justify your answer with appropriate working. 2
- c) If  $a+b=8$  and  $ab=17$ , find the value of  $\frac{2}{a} + \frac{2}{b}$  2
- d) Find the exact value of  $\cos 105^\circ$  2
- e) Prove  $\frac{\sin 3\theta}{\sin \theta} - \frac{\cos 3\theta}{\cos \theta} = 2$  ( $\sin \theta \neq 0$ ,  $\cos \theta \neq 0$ ) 3

**QUESTION 3 Start on a new page****Marks**

- a) Solve  $\frac{2x+1}{2x-1} \geq 2$  3
- b) Find the values of  $\tan x$  when  $\tan^2 x + \sec^2 x = 7$  2
- c) Fully simplify  $\sqrt{\frac{\sqrt{50}-\sqrt{32}}{\sqrt{8}}}$  without the aid of a calculator. Show all necessary working. 2
- d) Use the sine rule to find the value of  $\theta$  where  $\theta$  is obtuse 3

**QUESTION 4 Start on a new page**

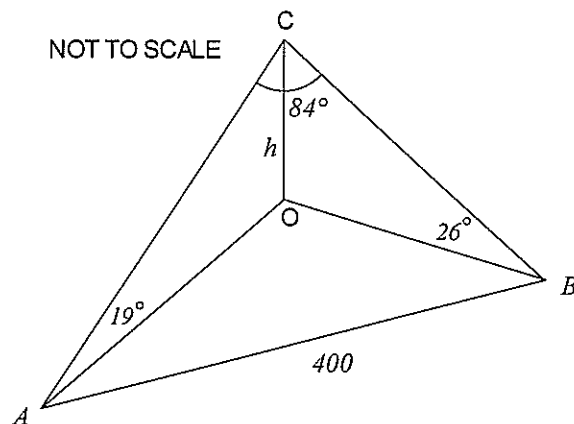
- a) Solve for  $0^\circ \leq x \leq 360^\circ$   
 $\cos 2x = \sin x$  3
- b) The sides of a triangle ABC are 16, 20, 24 cm



- i) Find the exact value of  $\cos A$  1
- ii) Hence, find the exact area of the triangle 2
- ii) Hence, find the distance from B to AC 2

**QUESTION 5 Start on a new page**

- a) Simplify  $\frac{5^n + 5^{n-2}}{5^{n-1}}$  2
- b) Factorise  $a^2 - b^2 + 2b - 1$  2
- c) A road runs by the side of a hill. From two points, A and B on the road, 400m apart, the angles of elevation of the top of the hill are  $19^\circ$  and  $26^\circ$  respectively. If the angle subtended by AB at the top of the hill is  $84^\circ$ , find the height of the hill to the nearest metre. 3



- d) If  $2\sin(\theta - 60^\circ) = \cos(\theta - 60^\circ)$  express  $\tan \theta$  in the form  $a + b\sqrt{3}$  where  $a$  and  $b$  are integers. 4

**QUESTION 6 Start on a new page**

- a) Prove that  $\frac{2 \tan A}{1 + \tan^2 A} = \sin 2A$  2
- b) i) Sketch the curves  $y = |x - 3|$  and  $y = 5 - |x|$ , on the same diagram, showing all necessary features. 2
- ii) Hence, or otherwise, solve  $|x - 3| + |x| < 5$  2
- c) Sketch the graph of  $y = \frac{x+2}{|x|}$  3
- d) If  $\cos x \cos y = \frac{1}{4}(\sqrt{3} - \sqrt{2})$  and  $\sin x \sin y = \frac{1}{4}(\sqrt{3} + \sqrt{2})$  find the smallest positive value of  $(x + y)$ . 3

**END OF EXAMINATION.**

1 a)  $(x+4)^2 = 3(x+4)$

$(x+4)(x+4-3) = 0$

$(x+4)(x+1) = 0$

$\therefore x = -4, -1$

① ↑ ②

b)  $\frac{1}{|x+5|} \geq 1 \quad (x \neq -5)$

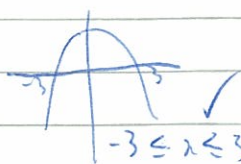
$1 \geq |x+5|$

$-1 \leq x+5 \leq 1$

$-6 \leq x \leq -4 \quad (x \neq -5) \checkmark$

c)  $9 - x^2 \geq 0 \quad x+1 > 0$

$x > -1$



$-3 \leq x \leq 3$

$x > -1$

Both exist when  $-1 < x \leq 3 \checkmark$

d)  $2^{2(2x+5)} = 2^3 \checkmark$

$\therefore 2x+5 = \frac{3}{2}$

$2x = -\frac{7}{2}$

$x = -\frac{7}{4} \checkmark$

e)  $y \geq 0$  and as  $\checkmark$

$x \rightarrow \infty \quad f(x) \rightarrow 1$

$x \rightarrow -\infty \quad f(x) \rightarrow 1$

$\therefore 0 \leq f(x) < 1 \checkmark$

2 a)  $2x+3y = 5$

$4x-3y = 24$

①  $4x+2y = 4$

③ - ②  $5y = -20$

$y = -4 \checkmark$

sub in ①  $2x-4 = 2$

$2x = 6$

$x = 3 \checkmark$

① for sub  
up of  
both eqns

5/b)  $a^2 - (b^2 - 2b + 1)$

$= a^2 - (b-1)^2 \checkmark$

$= (a+b-1)(a-(b-1))$

$= (a+b-1)(a-b+1) \checkmark$

(c)  $\frac{2}{a} + \frac{2}{b}$

$= \frac{2b+2a}{ab}$

$= \frac{2(a+b)}{ab} \checkmark$

$= \frac{2 \times 8}{17}$

$= \frac{16}{17} \checkmark$

d)  $\cos 105^\circ = \cos (60+45)^\circ \checkmark$

$= \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ$

$= \frac{1}{2} \times \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} \checkmark$

$= \frac{1-\sqrt{3}}{2\sqrt{2}} \text{ or } \frac{\sqrt{2}(1-\sqrt{3})}{4}$

e)  $\text{LHS} = \frac{\sin 3\theta \cos \theta}{\sin \theta \cos \theta} - \frac{\cos 3\theta \sin \theta}{\sin \theta \cos \theta} \checkmark$

$= \frac{\sin(3\theta - \theta)}{\sin \theta \cos \theta}$

$= \frac{\sin 2\theta}{\sin \theta \cos \theta} \checkmark$

$= \frac{2 \sin \theta \cos \theta}{\sin \theta \cos \theta}$

$= 2$

$= \text{RHS}$

$\therefore \frac{\sin 3\theta}{\sin \theta} = \frac{\cos 3\theta}{\cos \theta} = 2$

$$3(a) \frac{2n+1}{2n-1} \geq 2 \quad (n \neq \frac{1}{2})$$

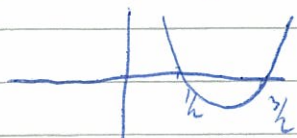
$$(2n+1)(2n-1) \geq 2(2n-1)^2$$

$$4n^2 - 1 \geq 2(4n^2 - 4n + 1)$$

$$4n^2 - 1 \geq 8n^2 - 8n + 2$$

$$0 \geq 4n^2 - 8n + 3$$

$$(2n-1)(2n-3) \leq 0$$



$$\frac{1}{2} \leq n \leq \frac{3}{2}$$

$$(b) \tan^2 n + \sec^2 n = 7$$

$$\tan^2 n + \tan^2 n + 1 = 7$$

$$2\tan^2 n = 6$$

$$\tan^2 n = 3$$

$$\tan n = \pm\sqrt{3}$$

$$(c) \sqrt{\frac{5\sqrt{2} - 4\sqrt{2}}{2\sqrt{2}}}$$

$$= \sqrt{\frac{1\sqrt{2}}{2\sqrt{2}}}$$

$$= \frac{1}{\sqrt{2}}$$

$$(d) \frac{\sin \theta}{\sqrt{2}} = \frac{\sin 45^\circ}{\sqrt{2}}$$

$$\sin \theta = \frac{2\sqrt{2}}{2\sqrt{2}} \cdot \frac{1}{\sqrt{2}}$$

$$\sin \theta = \frac{\sqrt{2}}{2}$$

$$\text{acute } \theta = 60^\circ$$

$$\theta = 180 - 60$$

$$= 120^\circ$$

$$4(a) 1 - 2\sin^2 n = \sin n$$

$$2\sin^2 n + \sin n - 1 = 0$$

$$(2\sin n - 1)(\sin n + 1) = 0$$

$$\sin n = \frac{1}{2} \quad \sin n = -1$$

$$n = 30^\circ, 150^\circ \quad n = 270^\circ$$

$$\therefore n = 30^\circ, 150^\circ, 270^\circ$$

$$(b) \cos A = \frac{20^2 + 24^2 - 16^2}{2 \times 20 \times 24}$$

$$\cos A = 0.75$$



$$\therefore \text{Area} = \frac{1}{2} \times 20 \times 24 \times \sin A$$

$$= 240 \times \frac{\sqrt{5}}{4}$$

$$= 60\sqrt{5}$$

$$(ii) \text{Area} = \frac{24 \times h}{2}$$

$$60\sqrt{5} = 12h$$

$$h = \frac{60\sqrt{5}}{12}$$

$$h = 5\sqrt{5}$$

$$5(a) \frac{5^n + 5^{n+2}}{5^{n+1}}$$

$$= \frac{5^{n+2}(5^2 + 1)}{5^{n+2} \times 5^1}$$

$$= \frac{26}{5}$$

$$\boxed{Q2(b)} f(-n) = (-n)^3 \cos(-n)$$

$$= -n^3 \cos n \quad \checkmark \text{ since } \cos(-n) = \cos n$$

$$= -f(n)$$

$\therefore$  function is odd

$$5(c) \sin 19^\circ = \frac{h}{AC}$$

$$h = AC \sin 19^\circ$$

$$\text{Also } \sin 26^\circ = \frac{h}{BC}$$

$$h = BC \sin 26^\circ$$

① expressions for both BC and AC

In  $\triangle ABC$

$$400^2 = \left(\frac{h}{\sin 26^\circ}\right)^2 + \left(\frac{h}{\sin 19^\circ}\right)^2 - \frac{2h \cdot h}{\sin 26^\circ \sin 19^\circ}$$

$$160000 = \frac{h^2}{\sin^2 26^\circ} + \frac{h^2}{\sin^2 19^\circ} - \frac{2 \cos 84^\circ h^2}{\sin 26^\circ \sin 19^\circ}$$

$$160000 h^2 \left( \frac{1}{\sin^2 26^\circ} + \frac{1}{\sin^2 19^\circ} - \frac{2 \cos 84^\circ}{\sin 26^\circ \sin 19^\circ} \right)$$

$$h^2 = \frac{160000}{\sec^2 26^\circ + \sec^2 19^\circ - 2 \cos 84^\circ \sec 26^\circ \sec 19^\circ}$$

$$h^2 = 12145.707 \dots$$

$$h = 110.207 \dots$$

$$h = 110 \text{ m}$$

$$6(a) 2 \sin(\theta - 60^\circ) = \cos(\theta - 60^\circ)$$

$$2 \tan(\theta - 60^\circ) = 1$$

$$\tan(\theta - 60^\circ) = \frac{1}{2}$$

$$\frac{\tan \theta - \tan 60^\circ}{1 - \tan \theta \tan 60^\circ} = \frac{1}{2}$$

$$\frac{\tan \theta - \sqrt{3}}{1 - \tan \theta \sqrt{3}} = \frac{1}{2}$$

$$2 \tan \theta - 2\sqrt{3} = 1 - \sqrt{3} \tan \theta$$

$$\tan \theta (2 + \sqrt{3}) = 1 + 2\sqrt{3}$$

$$\tan \theta = \frac{1 + 2\sqrt{3}}{2 + \sqrt{3}} \cdot \frac{2 - \sqrt{3}}{2 - \sqrt{3}}$$

$$= \frac{2 + 3\sqrt{3} - 6}{4 - 3}$$

$$\tan \theta = -4 + 3\sqrt{3}$$

6(a)

$$\text{LHS} = \frac{2 \sin A}{\cos A}$$

$$\frac{1 + \frac{\sin 2A}{\cos^2 A}}{\cos^2 A + \sin^2 A}$$

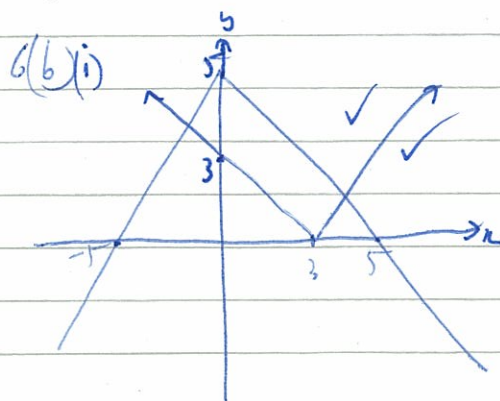
$$= \frac{2 \sin A \cos A}{\cos^2 A + \sin^2 A}$$

$$= \frac{\sin 2A}{1}$$

$$= \sin 2A$$

$$= \text{RHS}$$

$$\therefore \frac{2 \tan A}{1 + \tan^2 A} = \sin 2A$$



1 each for graphs with intercept  
or 1 for both graphs without intercept

$$6(b)(i) |x - 3| < 5 - |x|$$

Proof intersection:

$$-(x - 3) = 5 - (-x) \quad \& \quad x - 3 = 5 - x$$

$$-x + 3 = 5 + x \quad 2x = 8$$

$$-2 = 2x \quad x = 4$$

$$x = -1$$

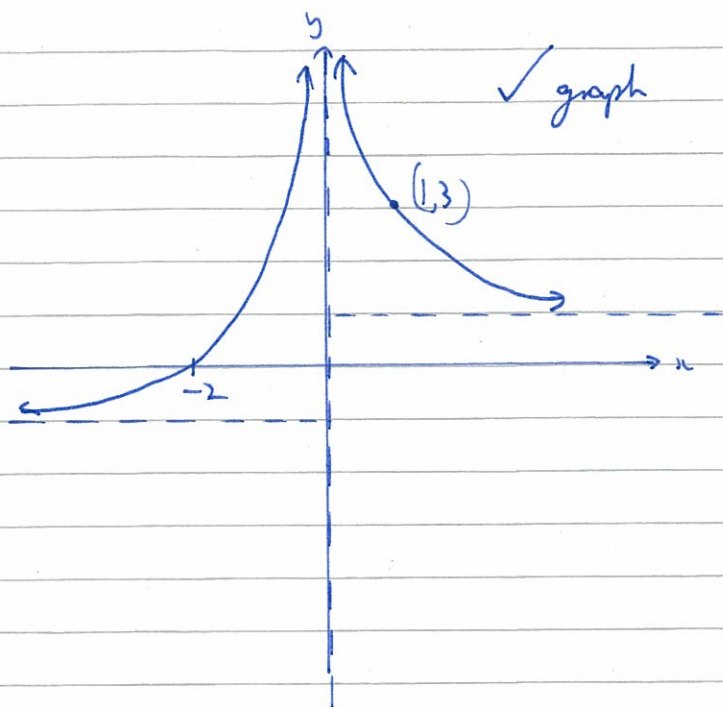
$\therefore -1 < x < 4$  is solution

$$6(c) \text{ If } x < 0$$

$$y = \frac{x+2}{-x} = -1 - \frac{2}{x}$$

$$\text{If } x > 0$$

$$y = \frac{x+2}{x} = 1 + \frac{2}{x}$$



$$\begin{aligned}
 6(d) \quad \cos(x+y) &= \cos x \cos y - \sin x \sin y \quad \checkmark \\
 &= \frac{1}{4}(\sqrt{3}-\sqrt{2}) - \frac{1}{4}(\sqrt{3}+\sqrt{2}) \\
 &= \frac{-2\sqrt{2}}{4}
 \end{aligned}$$

$$= -\frac{\sqrt{2}}{2}$$

$$= -\frac{1}{\sqrt{2}} \quad \checkmark$$

$$\begin{aligned}
 x+y &= \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) \\
 x+y &= 135^\circ \text{ is smallest positive value } \checkmark
 \end{aligned}$$